

Hands-On for C++

Vehicle Service Station

Aim

- To develop a console based application for a vehicle service station / garage.
- To provide various services to customers, prepare bills and track daily business by service station.

Technologies used

- C++ Programming:
 - Class & Objects
 - Inheritance & Polymorphism
 - Composition / Association
 - Operator overloading
 - File I/O
 - STL

Duration

- Object Oriented Design: 2 Hours
- Implementation: 14 Hours

Description

- Vehicle service station provide following services to customer:
 - Maintenance/Repairing:
 - Customer can ask to repair one or more areas e.g. dash board, brake system, etc.
 - Each repairing may need to change few parts e.g. dash board repairing may need to change speedo-meter cable, dash board dial, few indicators on dash board, etc.
 - For each repairing some labor charges might be applicable other than pricing of replaced spare parts.
 - Maintenance can also contain just fitting/tightening few screws, oiling of some parts for better performance; but not replacing any spare part. In this case only labor charges are applicable.
 - Engine/Gear Oil change:
 - Occasionally customer may ask or service engineer may advice to change engine and/or gear oil for better performance. Example, Servo4T, Castrol, etc. The engine oil is packaged in quantity of 900ml/1000ml (or any other quantity) for a fixed price.
 - Other than engine oil few fuel additives can be added for smoother performance of the vehicle. These additives are added into fuel tank. Like engine oil they are also sold in fixed quantity and fixed price.
 - As service station get reasonable margin in selling these oil packs, no separate charges are taken from customer for changing oil.
- For each servicing request from customer for a vehicle may including repairing of multiple parts and/or changing engine oil and/or adding additives.
- A bill is prepared for each service request separately.
 - The detailed bill should print following details:
 - Service station Name.
 - Customer Name and Vehicle Details for which services were provided.

- Service Date.
 - Maintenance Charges i.e. labor charges.
 - Each spare part costing
 - Oil/Additive price
 - Bill Amount + 12.6 % = Final Bill
- After customer paid bill amount, so balance amount. This balance can be added into next billing for that customer. (This feature is not implemented into given demo application).
- Service station keeps track of repeating customers and their vehicles (brought for servicing).
- The minimum information for any vehicle is its manufacturer, model and vehicle number (on number plate). Note that customer may have more than one vehicle.
- Also customer information should be maintained like his/her name, address, mobile number and vehicles. For sake of simplicity customers are identified by their name. However for proper implementation some ID proof information can be maintained.
- Service station should maintain daily billing details, so that at any time current cash collection can be calculated. The complete implementation of this case study may store all billing information for business analysis on weekly, monthly or yearly basis. (This feature is optional and not implemented into given demo application).
- All customer details should be loaded from file when application starts and store back to file when application terminate.
- As application starts a menu appears for service station manager:
 - New Customer
 - Input customer details and store it for future reference.
 - New Servicing Request (for Existing Customer)
 - Choose customer from existing list of customer.
 - Display his/her vehicle details.
 - Choose vehicle from the list or input details of new vehicle and store it for future reference.
 - Process service request i.e. provide maintenance, repairing and/or oil change services as per need.
 - Prepare bill and display it.
 - Get bill payment from customer and show balance.
 - Today's Business (Cash)
 - Compute total cash from list of all bills (paid amount) and display it.
- **Notes:**
 - Service station should maintain list of customers as List.
 - Service station should maintain list of bills as List.
 - Customer should contain list of vehicles as vector.
 - Each repairing will store list of replaced spare parts List.
 - Details of multiple services provided (during single service request) should be maintained as List.
 - Overload insertion/extraction operators to input details from user wherever appropriate.
 - You may use char array (instead of string type), so that file IO become easier task.